**Supplementary Table 2.** Publicly Available Databases

| Dataset | Record | Signal | Duration | Sampling rate (Hz) | Digitization | Annotation | Institution or Source |
|---|---|---|---|---|---|---|---|
| Apnea-ECG Database[116] (Apnea-ECG DB) | 35 train, 35 test | 1-lead ECG | 7–10 h | 100 | 16 bit resolution | Apnea annotations | Phillips-University, Marburg, Germany |
| ANSI/AAMI EC13 Test Waveforms[117] (ANSI/AAMI EC13) | N/A | 12-lead ECG | N/A | 720 | 12 bit resolution | N/A | N/A |
| European ST-T Database[118] (EST-TDB) | 90 | 2-lead ECG | 2 h | 250 | 12 bit resolution over a range of ±20 millivolts | Beat annotations | The CNR Institute for Clinical Physiology in Pisa |
| Heart for Heart[119] | 4769 individuals | PPG | 90s | 30 | N/A | Sex, age, weight, height, and smoking | Smartphone happitech application |
| Long-Term Atrial Fibrillation Database[120] (LTAFDB) | 84 | 2-lead ECG | 24–25 h | 128 | 12 bit resolution over a 20 mV range | Beat annotations | Boston's Beth Israel Deaconess Medical Center (Boston's Beth Israel Hospital) |
| Long Term ST Database[121] (LTSTDB) | 86 | 2,3-lead ECG | 21–24 h | 250 | 12 bit resolution over a range of ±10 millivolts | Beat annotations | Ambulatory electrocardiographic recorders |
| MIT-BIH Arrhythmia Database[122] (MITADB) | 48 | 2-lead ECG | 30 m | 360 | 11 bit resolution over a 10 mV range | Beat annotations | Del Mar Avionics model 445 |
| MIT-BIH Atrial Fibrillation Database[123] (MITAFDB) | 23 | 2-lead ECG | 10 h | 250 | 12 bit resolution over a range of ±10 millivolts | Beat annotations | Boston's Beth Israel Deaconess Medical Center (Boston's Beth Israel Hospital) |
| MIT-BIH Long-Term Database[117] (MITLTDB) | 7 | ECG | 14–22 h | N/A | N/A | Beat annotations | N/A |
| MIT-BIH Normal Sinus Rhythm Database[117] (MITLTDB) | 18 | 2-lead ECG | N/A | N/A | N/A | N/A | Boston's Beth Israel Deaconess Medical Center (Boston's Beth Israel Hospital) |
| MIT-BIH Malignant Ventricular Arrhythmia Database[117] (MITMVADB) | 22 | 2-lead ECG | 30 m | N/A | N/A | Beat annotations | N/A |
| MIT-BIH Noise Stress Test Database[124] (MITNST) | 15 | 2-lead ECG | 3 h, 12 h | 360 | N/A | Beat annotations | By the script nstdbgen- using two clean recordings (118 and 119) from the MIT-BIH Arrhythmia Database |
| Multiparameter Intelligent Monitoring in Intensive Care Database II[125] (MIMIC II) | 26000 intensive care unit patients | PPG | N/A | 125 | N/A | Complex-parameter Annotations | Bedside workstations as well as hospital archives |
| Multiparameter Intelligent Monitoring in Intensive Care Database III[126] (MIMIC III) | 53423 | PPG | N/A | 125 | 12 bit | Complex-parameter Annotations | Critical care units of the Beth Israel Deaconess Medical Center between 2001 and 2012 |
| Normal Sinus Rhythm RR Interval Database[117] (NSR-RR) | 54 | ECG | N/A | 128 | N/A | Beat annotations | Washington University School of Medicine, St. Louis, Columbia-Presbyterian Medical Center, New York |
| PTB Diagnostic ECG Database[135] (PTB diagnostic ECG DB) | 549 records from 290 subjects | 16 input channels (14 for ECGs, 1 for respiration, 1 for line voltage) | N/A | 1000 | 16 bit resolution over a range of ±16.384 mV | Detailed clinical summary | Physikalisch-Technische Bundesanstalt, the National Metrology Institute of Germany |
| PTB-XL[136] (PTB-XL) | 21837 | 12-lead ECG | 10 s | 500 | 16 bit precision at a resolution of 1 μV/LSB | Disease annotation | Devices from Schiller AG |
| SOMNIA and HealthBed Datasets[127] (SOMNIA and HealthBed DB) | 2100 subjects | Polysomnography | One night | The suprasternal pressure signals: 128 | 24 bit version of the 16-bit European Data Format 'plus' | Related sleep phenomena | Kempenhaeghe Sleep Center, Polysomnography sensors |
| The Beth Israel Deaconess Medical Center Congestive Heart Failure Database[128] (BIDMC-CHF) | 15 | 2-lead ECG | 20 h | 250 | 12 bit resolution over a range of ±10 millivolts | N/A | Boston's Beth Israel Deaconess Medical Center (Boston's Beth Israel Hospital) |
| The Chapman ECG Database[137] (Chapman ECG DB) | 10,605 | 12-lead ECG | 10 s | 500 | 32 bit | Beat annotations | The GE MUSE ECG system |
| The China Physiological Signal Challenge 2020 Database[129] (CPSCDB) | 10 train, 10 test | 1-lead ECG | 24 h | 400 | N/A | Beat annotations | Wearable ECG device |
| The Congestive Heart Failure RR interval Database[117] (CHF-RR) | 29 | ECG | N/A | 128 | N/A | Beat annotations | Columbia-Presbyterian Medical Center, New York |
| The Fantasia Database[130] (FD) | 20 youngs, 20 elders | 1-lead ECG | 2 h | 250 | N/A | Beat annotations | N/A |
| The VitalDB[131] | 561150 (6423 surgical patients) | N/A | N/A | 250 | N/A | Demographic data and surgery information | Seoul National University Hospital, Seoul, Republic of Korea (Vital recorder program) |
| The PhysioNet Computing in Cardiology Challenge 2017[117] (PhysioNet/ CinC Challenge 2017 DB) | 8528 train, 3658 test | 1-lead ECG | 9–60 s | 300 | N/A | Beat annotations | AliveCor device |
| The PhysioNet Atrial Fibrillation Prediction Database[132] (AFPDB) | 50 train, 50 test | 2-lead ECG | 30 m | 128 | 16 bit resolution | Beat annotations | Steven Swiryn of Northwestern University |
| The Public Telehealth Database[133] (THDB) | 300 | 1 | N/A | 500 | 12 bit resolution over a 5 mV range | Beat annotations | TeleMedCare Health Monitor—TMC-HM (TeleMedCare Pty Ltd Sydney, Australia) |
| The UCD Overnight Database[117] (UCDODB) | 25 | 3-lead ECG | Overnight | N/A | N/A | Sleep stages | St. Vincent's University Hospital Sleep Disorders Clinic |
| 2015 IEEE Signal Processing Cup Database[134] | 12 | ECG & PPG | 0.5, 1 min | 125 | N/A | N/A | Pulse oximeter with green LED, Wet ECG sensors |

ECG, electrocardiography; PPG, photoplethysmography.